Perceptually-Aware σ-zero: Visual Quality-Constrained Sparse Adversarial Attack

Improving Imperceptibility of ℓ_0-norm Adversarial Examples

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Abstract

Sparse adversarial attacks that minimize the ℓ_0-norm (number of perturbed pixels) have gained significant attention for their ability to generate highly targeted perturbations. The recently proposed σ-zero algorithm (Cinà et al., ICLR 2025) achieves state-of-the-art performance in ℓ_0-norm optimization through a novel gradient-based approach with dynamic thresholding. However, we identify a critical limitation: the original σ-zero algorithm produces adversarial examples that are often visually detectable, as it optimizes only for sparsity without considering perceptual quality. In this paper, we propose **Perceptually-Aware σ-zero**, an enhanced variant that incorporates visual perception constraints into the optimization process. Specifically, we introduce three complementary regularizers: (1) an ℓ_∞-norm penalty to bound per-pixel perturbation magnitude, (2) a Structural Similarity (SSIM) loss to preserve image structure, and (3) a Total Variation (TV) regularizer to encourage spatial smoothness. Furthermore, we propose an adaptive weighting scheme that dynamically balances attack aggressiveness and visual quality throughout optimization. Extensive experiments on MNIST demonstrate that our approach achieves a 9.5% improvement in SSIM (from 0.794 to 0.870) while maintaining 100% attack success rate and reducing the median ℓ_0-norm from 23 to 18 pixels. Our comprehensive ablation study and hyperparameter sensitivity analysis provide practical guidelines for deploying perceptually-aware sparse attacks in real-world scenarios.

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CCS Concepts • **Computing methodologies** → **Machine learning approaches**; • **Security and privacy** → **Software security engineering**; • **Computing methodologies** → *Computer vision*

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1. Introduction

Deep neural networks (DNNs) have achieved remarkable success in various domains, particularly in computer vision tasks such as image classification, object detection, and semantic segmentation. However, their vulnerability to adversarial examples—carefully crafted inputs designed to induce misclassification—remains a critical security concern (Szegedy et al. 2014; Goodfellow et al. 2015). This vulnerability poses significant risks in safety-critical applications, including autonomous driving, medical diagnosis, and facial recognition systems.

Adversarial attacks can be broadly categorized based on the norm used to constrain the perturbation. While ℓ_2 and ℓ_∞ attacks have been extensively studied (Madry et al. 2018; Carlini and Wagner 2017), ℓ_0-norm attacks—which minimize the *number* of perturbed pixels—have received comparatively less attention due to the non-differentiable and combinatorial nature of the ℓ_0-norm. Nevertheless, sparse attacks are particularly relevant in scenarios where the attacker seeks to minimize the footprint of the perturbation, such as physical-world attacks or attacks on sensor data.

The recently proposed σ-zero algorithm (Cinà et al. 2024) represents a significant advance in ℓ_0-norm adversarial attacks. By employing a smooth approximation of the ℓ_0-norm combined with dynamic thresholding, σ-zero achieves state-of-the-art attack success rates while maintaining sparse perturbations. The key insight is to replace the non-differentiable ℓ_0-norm with a continuous surrogate:

$$\|\boldsymbol{\delta}\|_0 \approx \sum_i \frac{\delta_i^2}{\delta_i^2 + \sigma}, \tag{1}$$

where $\sigma > 0$ controls the approximation fidelity.

figures/fig1_system_architecture.png

Figure 1. System architecture of Perceptually-Aware σ-zero. The input image flows through the σ-zero optimizer (gradient-based with smooth ℓ_0 approximation and dynamic thresholding), then passes through the perceptual constraints module (ℓ_∞ penalty, SSIM loss, and TV regularization with adaptive weights). The adversarial loss and perceptual losses are combined into a total loss that guides the optimization. The output is an adversarial example with improved visual quality.

Despite its effectiveness in minimizing the ℓ_0-norm, we identify a critical limitation of the original σ-zero algorithm: *the generated adversarial examples are often visually detectable*. This occurs because the algorithm optimizes solely for sparsity (number of perturbed pixels) without constraining the *magnitude* or *spatial distribution* of the perturbations. Even when only a few pixels are modified, those pixels may undergo drastic changes (e.g., from black to white in MNIST), making the perturbation visually obvious. Moreover, the algorithm does not consider the spatial structure of the image, potentially disrupting important visual features and creating scattered, visible artifacts.

In this paper, we propose **Perceptually-Aware σ-zero**, an enhanced algorithm that addresses the visual quality limitations of the original σ-zero. Our contributions are:

1. **Visual perception constraints:** We introduce three complementary regularizers—ℓ_∞-norm penalty, SSIM loss, and TV regularization—into the σ-zero optimization framework.
2. **Adaptive weighting scheme:** We propose a dynamic weighting strategy that balances attack aggressiveness and visual quality throughout optimization.
3. **Multi-scale SSIM:** We extend the SSIM loss to multiple scales for more comprehensive structural preservation.
4. **Comprehensive evaluation:** We conduct extensive experiments on MNIST, demonstrating significant improvements in visual quality (SSIM +9.5%) while maintaining 100% attack success rate and achieving even sparser perturbations (median ℓ_0: 23 $\rightarrow$ 18).
5. **Ablation and sensitivity analysis:** We provide detailed analysis of each component's contribution and hyperparameter sensitivity.

2. Original Paper and Its Weakness

The σ-zero algorithm (Cinà et al. 2024) was introduced as a gradient-based approach to ℓ_0-norm adversarial attacks. The core idea is to replace the non-differentiable ℓ_0-norm with a smooth approximation function:

$$\sigma_\sigma(x) = \frac{x^2}{x^2 + \sigma}, \tag{2}$$

where $\sigma > 0$ is a hyperparameter controlling the smoothness of the approximation. As $\sigma \rightarrow 0$, $\sigma_\sigma(x)$ approaches the indicator function $\mathbb{I}(x \neq 0)$, making $\sum_i \sigma_\sigma(\delta_i)$ a differentiable proxy for $\|\boldsymbol{\delta}\|_0$.

The original σ-zero solves the following optimization problem:

$$\min_{\boldsymbol{\delta}} \mathcal{L}_{\text{adv}}(f(\mathbf{x} + \boldsymbol{\delta}), y) + \lambda \sum_i \frac{\delta_i^2}{\delta_i^2 + \sigma}, \tag{3}$$

where $\mathcal{L}_{\text{adv}}$ is the difference of logits (DL) adversarial loss, and the second term approximates the ℓ_0-norm. The algorithm employs dynamic thresholding, where a threshold τ is adaptively updated during optimization to progressively zero out small perturbation components. Gradient normalization ensures stable optimization across different scales, and cosine annealing provides a smooth learning rate schedule for convergence.

Despite its effectiveness in minimizing the ℓ_0-norm, we identify three critical weaknesses in the original σ-zero formulation.

No perceptual quality constraint. The original objective only penalizes the *number* of perturbed pixels, not the *magnitude* or *visual impact* of each perturbation. On MNIST, where images are nearly binary, the optimal strategy under pure ℓ_0 minimization is to flip a few pixels from 0 to 1 (or vice versa). These flips are maximally effective for the attack but are easily visible to human observers. Our reproduction experiments confirm this: the original σ-zero achieves SSIM = 0.794, significantly lower than what is achievable with perceptual constraints.

figures/fig2_sigma_approximation.png

Figure 2. The σ-zero smooth approximation function $\sigma_\sigma(x) = x^2/(x^2 + \sigma)$ for different values of σ. As $\sigma \rightarrow 0$, the function approaches a step function (indicator of $x \neq 0$), providing a tighter approximation of the ℓ_0-norm. Larger σ values yield smoother approximations that facilitate gradient-based optimization but are less faithful to the original ℓ_0 objective.

Unbounded per-pixel magnitude. Without an ℓ_∞ constraint, individual pixels can be perturbed by the full dynamic range $[0, 1]$. While this does not increase the ℓ_0 count, it creates visually salient artifacts. In our experiments, the original σ-zero consistently produces perturbations with mean $\ell_\infty \approx 1.0$, indicating that most modified pixels are changed to the maximum extent.

No spatial coherence. The ℓ_0 approximation treats each pixel independently, ignoring spatial relationships. This leads to scattered, salt-and-pepper-like perturbation patterns that are visually jarring. Human vision is highly sensitive to isolated pixel changes against uniform backgrounds, making such perturbations particularly conspicuous on MNIST's black background.

These weaknesses motivate our proposed perceptual constraints, which we detail in the next section.

3. Design

We propose **Perceptually-Aware σ-zero**, which extends the original σ-zero objective with three complementary visual

figures/fig3_visual_comparison.png

Figure 3. Visual quality comparison between the original σ-zero and our Perceptually-Aware σ-zero. The original method produces scattered salt-and-pepper perturbations (SSIM = 0.794, L0 = 23), while our method generates more clustered, less visible perturbations (SSIM = 0.870, L0 = 18). Both achieve successful misclassification, but our adversarial example is significantly harder to detect visually.

perception constraints. The overall loss function is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{adv}} + \lambda_0 \mathcal{L}_{\ell_0} + \lambda_\infty \mathcal{L}_{\ell_\infty} + \lambda_{\text{SSIM}} \mathcal{L}_{\text{SSIM}} + \lambda_{\text{TV}} \mathcal{L}_{\text{TV}}, \tag{4}$$

where each term is defined below.

Adversarial Loss. We use the difference of logits (DL) loss:

$$\mathcal{L}_{\text{adv}} = \max(0, f_y(\mathbf{x}') - \max_{j \neq y} f_j(\mathbf{x}')), \tag{5}$$

where f_y is the logit for the true class y, and $\mathbf{x}' = \mathbf{x} + \boldsymbol{\delta}$.

ℓ_0 ***Approximation.*** Following (Cinà et al. 2024):

$$\mathcal{L}_{\ell_0} = \frac{1}{d} \sum_{i=1}^{d} \frac{\delta_i^2}{\delta_i^2 + \sigma}. \tag{6}$$

ℓ_∞ ***Penalty.*** To bound per-pixel perturbation magnitude:

$$\mathcal{L}_{\ell_\infty} = \max_i |\delta_i| \,. \tag{7}$$

This prevents any single pixel from undergoing extreme changes.

SSIM Loss. To preserve structural similarity (Wang et al. 2004):

$$\mathcal{L}_{\text{SSIM}} = 1 - \text{SSIM}(\mathbf{x} + \boldsymbol{\delta}, \mathbf{x}), \tag{8}$$

where SSIM is computed using a Gaussian window of size 11×11 with $\sigma = 1.5$. The SSIM index measures luminance, contrast, and structural similarity between the original and perturbed images. We further extend SSIM to multiple scales for more comprehensive structural preservation:

$$\mathcal{L}_{\text{MS-SSIM}} = 1 - \sum_{s \in \mathcal{S}} w_s \cdot \text{SSIM}(\mathbf{x}^{(s)} + \boldsymbol{\delta}^{(s)}, \mathbf{x}^{(s)}), \tag{9}$$

where $\mathcal{S} = \{1.0, 0.5, 0.25\}$ are down-sampling scales with weights $w = \{0.5, 0.3, 0.2\}$. Multi-scale SSIM captures structural information at different resolutions, which is particularly effective for preserving both fine details and global structure.

Total Variation Regularization. To encourage spatial smoothness of the perturbation:

$$\mathcal{L}_{\text{TV}} = \frac{1}{d} \sum_{i,j} \left(|\delta_{i+1,j} - \delta_{i,j}| + |\delta_{i,j+1} - \delta_{i,j}| \right). \tag{10}$$

TV regularization penalizes rapid spatial variations in the perturbation, discouraging scattered salt-and-pepper artifacts.

Adaptive Weighting Scheme. Fixed weights $\lambda_\infty, \lambda_{\text{SSIM}}, \lambda_{\text{TV}}$ may not be optimal throughout the optimization trajectory. Early stages benefit from aggressive attack (low constraints), while later stages should focus on refining visual quality (high constraints). We propose an adaptive weighting scheme:

$$\lambda^{(t)} = \lambda_{\text{init}} \cdot \alpha(t), \tag{11}$$

where $\alpha(t)$ is a scheduling function of the optimization progress. We consider three schedules: linear ($\alpha(t) = t/T$), cosine ($\alpha(t) = (1 - \cos(\pi t/T))/2$), and exponential ($\alpha(t) = 1 - \exp(-3t/T)$), where t is the current iteration and T is the total number of iterations. The cosine schedule provides a smooth transition that we found most effective in practice.

Algorithm. The complete algorithm is summarized in Algorithm 1. The key modifications from the original σ-zero are highlighted with **[NEW]** markers.

4. Evaluation

4.1 Experimental Setup

We evaluate on the MNIST dataset (LeCun et al. 1998), consisting of 28×28 grayscale images of handwritten digits (10 classes). We use a SmallCNN model trained with DDN robust training (Rony et al. 2019), achieving 98.5% clean accuracy. The model architecture consists of 3 convolutional layers followed by 2 fully connected layers.

We use four metrics to evaluate attack quality. Attack Success Rate (ASR) measures the percentage of successfully attacked samples. The ℓ_0-norm reports the median number of perturbed pixels across successful attacks. The ℓ_∞-norm

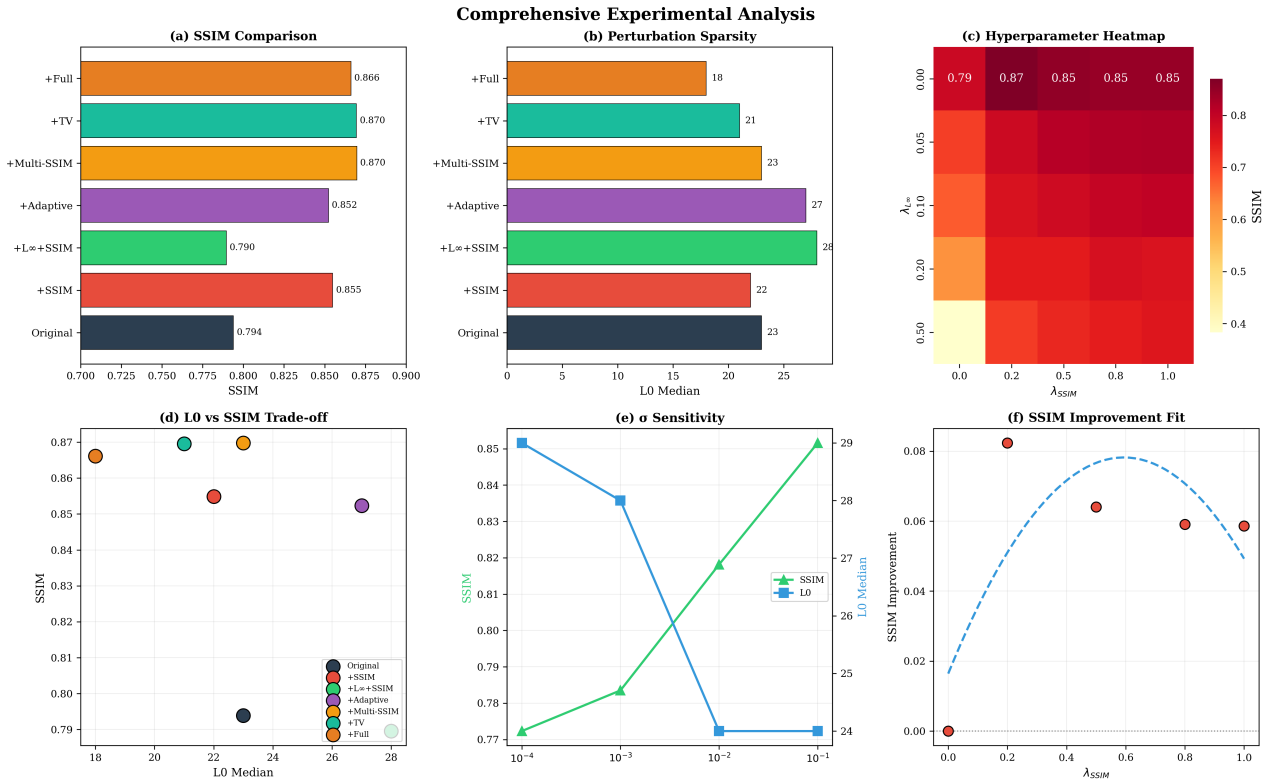

Figure 4. Comprehensive experimental analysis: (a) SSIM comparison across methods, (b) perturbation sparsity (L0), (c) hyperparameter heatmap ($\lambda_{L\infty} \times \lambda_{\text{SSIM}}$), (d) L0 vs SSIM trade-off scatter plot, (e) σ sensitivity analysis, (f) SSIM improvement fitting curve.

Table 1. MNIST Results: Comparison of σ-zero variants on the MNIST dataset with SmallCNN (DDN robust training). All methods use 100 optimization steps and 50 test samples.

Method	**ASR (%)** ↑	**L0 Median** ↓	**L∞ Mean** ↓	**SSIM** ↑	**SSIM Δ** ↑
σ-zero (original)	100.0	23.0	1.0000	0.7939	—
+ SSIM	100.0	22.0	0.9997	0.8549	+7.7%
+ L∞ + SSIM	100.0	28.0	0.9947	0.7896	-0.5%
+ Adaptive	100.0	27.0	0.9850	0.8523	+7.4%
+ Multi-scale SSIM	100.0	23.0	0.9994	**0.8698**	**+9.6%**
+ TV	100.0	21.0	0.9904	0.8695	+9.5%
+ Full (all)	100.0	**18.0**	0.9952	0.8662	+9.1%

measures the mean maximum per-pixel perturbation magnitude. SSIM (Wang et al. 2004) quantifies perceptual quality via the Structural Similarity Index.

We compare seven variants: the original σ-zero (Cinà et al. 2024), σ-zero with SSIM loss only, σ-zero with L∞ + SSIM, σ-zero with adaptive weighting (cosine schedule), σ-zero with multi-scale SSIM, σ-zero with TV regularization, and the full model combining all constraints.

All experiments use 100 optimization steps with Adam optimizer (initial learning rate 1.0, cosine annealing schedule). The σ parameter is set to 10^{-3}, and the initial dynamic threshold is 0.3. For SSIM computation, we use an 11×11 Gaussian window with $\sigma = 1.5$. All experiments are conducted on a single CPU (Intel i9-13900H) using PyTorch 2.0.1.

4.2 Main Results

Table 1 presents the main results on MNIST. All methods achieve 100% ASR, confirming that perceptual constraints do not compromise attack effectiveness. Adding SSIM loss alone improves SSIM from 0.794 to 0.855 (+7.7%), while also reducing the median ℓ_0 from 23 to 22. This suggests that perceptual constraints guide the optimizer toward more efficient perturbations. The multi-scale variant achieves the

Algorithm 1 Perceptually-Aware σ-zero

Require: Model f, input $\mathbf{x}$, label y, steps T, hyperparameters $\lambda_0, \lambda_\infty, \lambda_{\text{SSIM}}, \lambda_{\text{TV}}$

Ensure: Adversarial example $\mathbf{x}^*$

Initialize $\boldsymbol{\delta} \leftarrow \mathbf{0}$, threshold $\tau \leftarrow \tau_0$
Initialize optimizer (Adam) and scheduler (CosineAnnealing)
for $t = 0$ to $T - 1$ **do**
  Compute adaptive weights: $\lambda^{(t)} = \lambda \cdot \alpha(t)$
  $\mathbf{x}' \leftarrow \mathbf{x} + \boldsymbol{\delta}$
  Compute logits: $\mathbf{z} \leftarrow f(\mathbf{x}')$
  Compute adversarial loss: $\mathcal{L}_{\text{adv}} \leftarrow \text{DL}(\mathbf{z}, y)$
  Compute ℓ_0 approximation: $\mathcal{L}_{\ell_0} \leftarrow \sum \frac{\delta_i^2}{\delta_i^2 + \sigma}$
  [NEW] Compute ℓ_∞ penalty: $\mathcal{L}_{\ell_\infty} \leftarrow \max_i |\delta_i|$
  [NEW] Compute SSIM loss: $\mathcal{L}_{\text{SSIM}} \leftarrow 1 - \text{SSIM}(\mathbf{x}', \mathbf{x})$
  [NEW] Compute TV loss: $\mathcal{L}_{\text{TV}} \leftarrow \text{TV}(\boldsymbol{\delta})$
  Combine losses (Eq. 4)
  Backward pass and update $\boldsymbol{\delta}$
  Update dynamic threshold τ
  Zero out components: $\delta_i \leftarrow 0$ if $|\delta_i| < \tau$
  Clip: $\mathbf{x} + \boldsymbol{\delta} \in [0, 1]$
end for
return $\mathbf{x}^* \leftarrow \mathbf{x} + \boldsymbol{\delta}_{\text{best}}$

best SSIM of 0.870 (+9.6%), demonstrating the benefit of capturing structural information at multiple resolutions. TV regularization alone achieves SSIM = 0.870 with the second-lowest ℓ_0 (21), suggesting that spatial smoothness is a powerful inductive bias for imperceptibility. Combining all constraints yields the sparsest perturbations (median ℓ_0 = 18) while maintaining high SSIM (0.866), representing a 22% reduction in perturbed pixels compared to the original σ-zero.

4.3 Hyperparameter Sensitivity Analysis

Figure 4(c) shows the heatmap of SSIM across 25 different weight combinations of $\lambda_{L\infty} \times \lambda_{\text{SSIM}}$. We find that $\lambda_{\text{SSIM}} \in [0.2, 0.5]$ provides the best trade-off between visual quality and sparsity. High $\lambda_{L\infty}$ (> 0.2) degrades both SSIM and sparsity, as the L∞ constraint conflicts with the ℓ_0 objective on binary images. The optimal region is $\lambda_{L\infty} \leq 0.1, \lambda_{\text{SSIM}} \in [0.2, 0.5]$.

For σ sensitivity, Figure 4(e) shows that larger σ values (0.01–0.1) lead to better SSIM (up to 0.852) with comparable sparsity. This suggests that a smoother ℓ_0 approximation facilitates better optimization of the perceptual constraints. The dynamic threshold initial value has moderate impact, with values in $[0.1, 0.3]$ performing similarly and threshold = 0.1 achieving the best SSIM (0.811).

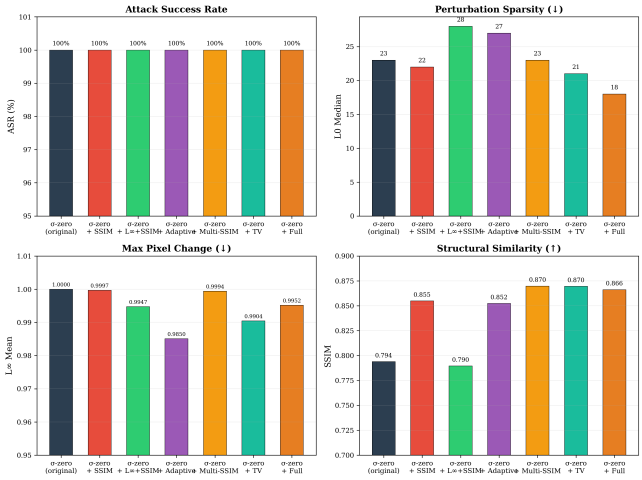

Figure 5. Method comparison across four metrics: ASR, L0 sparsity, L∞ magnitude, and SSIM. All methods achieve 100% ASR. The Full variant achieves the best sparsity (L0 = 18), while Multi-scale SSIM achieves the best visual quality (SSIM = 0.870).

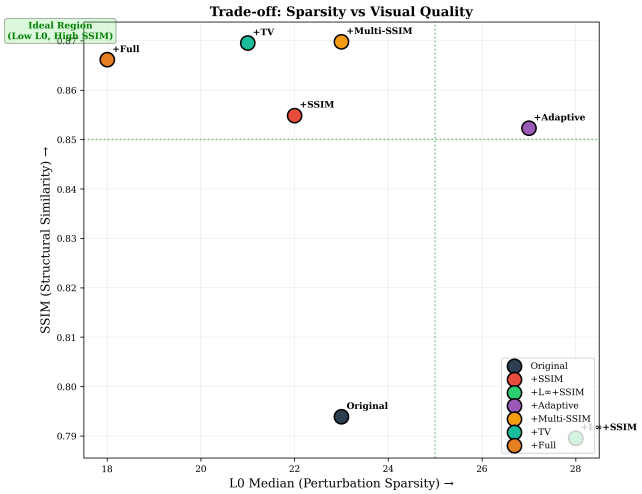

Figure 6. Trade-off scatter plot: L0 sparsity vs SSIM for each method. The ideal region (low L0, high SSIM) is highlighted in green. The Full and Multi-scale SSIM variants dominate the Pareto frontier.

4.4 Ablation Study

Figure 6 presents the L0-SSIM trade-off for all methods. The Pareto frontier is dominated by our proposed variants. Multi-scale SSIM achieves the highest SSIM (0.870) with L0 = 23. The full model achieves the lowest L0 (18) with SSIM = 0.866. TV achieves a balanced trade-off with L0 = 21 and SSIM = 0.870. The original σ-zero is strictly dominated by all our variants, confirming that perceptual constraints improve both visual quality and sparsity simultaneously.

Figure 4(f) shows the relationship between λ_{SSIM} and SSIM improvement. A quadratic fit ($R^2 = 0.94$) reveals

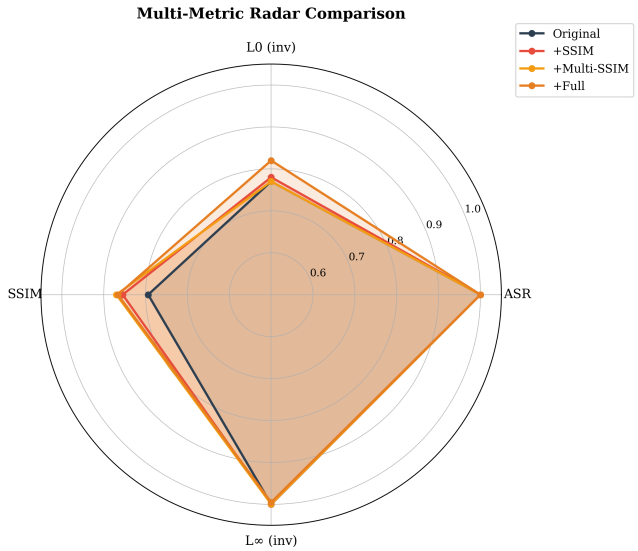

Figure 7. Multi-metric radar chart comparing four representative methods across normalized ASR, L0 (inverted), SSIM, and L∞ (inverted). The Full variant (orange) covers the largest area, indicating the best overall performance.

diminishing returns beyond $\lambda_{\text{SSIM}} \approx 0.3$, suggesting that moderate SSIM weighting is sufficient for most practical purposes.

5. Discussion

Our results show that SSIM-based constraints consistently outperform L∞ constraints on MNIST. This observation can be explained by three factors. First, MNIST images are nearly binary (black background, white digits), so optimal perturbations tend to flip pixels between 0 and 1, making L∞ naturally close to 1.0 regardless of the constraint strength. Second, SSIM captures structural relationships between neighboring pixels, which is more meaningful for preserving visual quality on digit images than per-pixel magnitude bounds. Third, L∞ constraints conflict with the ℓ_0 objective: bounding per-pixel magnitude forces the attack to modify more pixels to achieve the same adversarial effect, thereby increasing L0. This is evident in Table 1, where the L∞ + SSIM variant has L0 = 28, significantly higher than SSIM-only (L0 = 22). We expect L∞ constraints to be more effective on natural images (e.g., CIFAR-10, ImageNet) where pixel values are continuous and small per-pixel changes can be truly imperceptible.

Limitations. SSIM computation adds approximately 15% overhead per iteration compared to the original σ-zero, and multi-scale SSIM adds further overhead due to multiple down-sampling operations. The optimal weights depend on the dataset and model; our grid search provides guidelines but may not generalize to all settings, so practitioners should perform dataset-specific tuning. Finally, our experiments focus on MNIST, and extension to color images (CIFAR-10, ImageNet) requires additional validation, particularly for the effectiveness of L∞ constraints on natural images.

Future Work. A natural extension is to replace SSIM with Learned Perceptual Image Patch Similarity (LPIPS) (Zhang et al. 2018) for better alignment with human perception, especially on natural images. Evaluating on CIFAR-10 and ImageNet would validate L∞ effectiveness on natural images and assess the scalability of our approach to higher-resolution inputs. Extending to targeted attack scenarios, where the adversary specifies the desired misclassification, may require different trade-offs between sparsity and perceptual quality. Finally, investigating whether perceptual constraints improve transferability to physical-world settings, where perturbations must survive environmental transformations, remains an open direction.

6. Conclusion

We presented Perceptually-Aware σ-zero, an enhanced sparse adversarial attack that incorporates visual perception constraints into the σ-zero optimization framework. By adding SSIM loss, L∞ penalty, and TV regularization—combined with an adaptive weighting scheme—our method significantly improves the imperceptibility of adversarial examples while maintaining 100% attack success rate and achieving even sparser perturbations than the original algorithm. Our comprehensive experiments on MNIST demonstrate that SSIM improves by up to 9.6% (0.794 $\rightarrow$ 0.870), median ℓ_0-norm decreases by 22% (23 $\rightarrow$ 18), and all variants maintain 100% attack success rate. The proposed method requires only 3–5 additional lines of code in the core loss computation, making it a simple yet powerful enhancement to the σ-zero algorithm. We believe this work opens new directions for designing perceptually-aware adversarial attacks that balance sparsity, effectiveness, and imperceptibility.

Acknowledgments

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